

A

Project Review Meeting

on

E-Commerce Product Catalogue

By

Rishi Varma K
V Ashish Reddy

roll 2520030248
roll 2520030415

Team Number: 1

Under the guidance of

D. Pavithra
Associate Professor

Department of Computer Science and Engineering



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Outline of the Presentation

- Abstract
- Introduction
- Problem Statement
- Project Flow Diagram
- Objectives with Justification
- Modules[Frontend web pages]
- Expected Outcomes of the Proposed Work
- Conclusion
- References

Abstract

- The E-Commerce Product Catalogue is a web application that helps users browse, search, filter, and compare products efficiently. Developed using React.js, Node.js, Express.js, and MySQL, the system provides a user-friendly shopping experience while allowing administrators to manage products and categories through a dedicated dashboard.

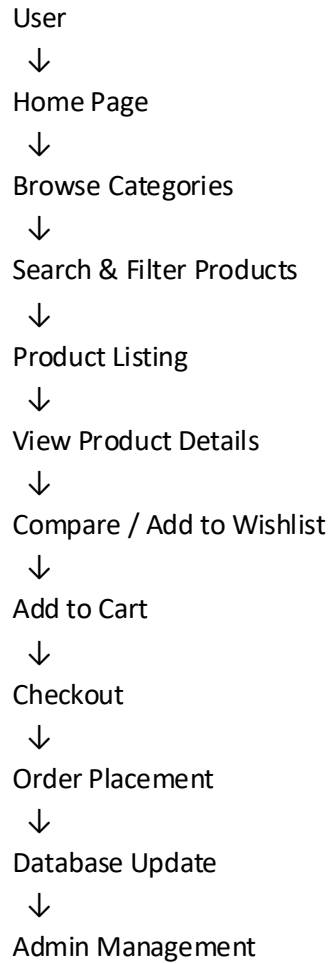
Introduction

- Online shopping platforms contain large numbers of products, making product discovery an important challenge. This project provides an organized catalogue system that enables users to find products easily through category navigation, filtering options, and search functionality while demonstrating modern full-stack web development practices

Problem Statement

- Customers often struggle to find relevant products because of poor navigation and limited filtering options. This project addresses the issue by providing category-based browsing, advanced search, product comparison, wishlist management, and an admin panel to improve both user experience and system management.

Project Flow Diagram



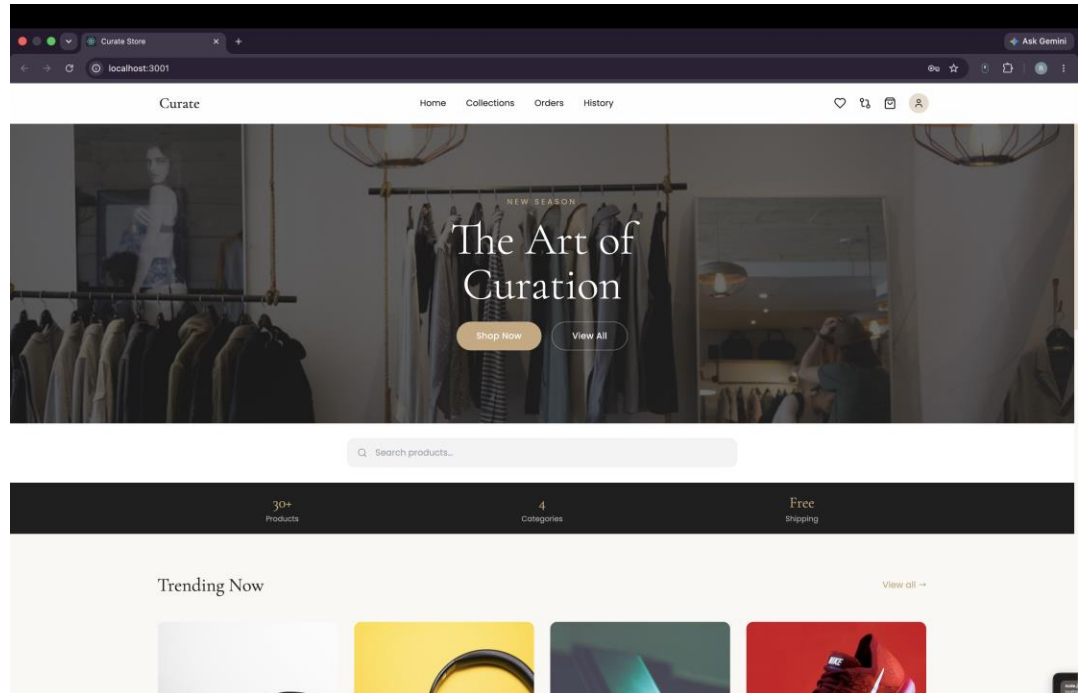
Objectives with Justification

- **1. Multi-Level Category Navigator**
 - Justification: Hierarchical categories help users drill to specific types quickly.
- **2. Rich Filter Panel (Price, Brand, Rating, Colour, Size)**
 - Justification: Granular filters reduce irrelevant results and improve conversion.
- **3. Real-Time Search with Auto-Suggestions**
 - Justification: Reduces typing effort and guides users to valid product queries.
- **4. Product Comparison Tool**
 - Justification: Side-by-side comparison enables informed purchase decisions.
- **5. Wishlist & Browsing History**
 - Justification: Personalization increases engagement and return visit rates.
- **6. Admin Panel**
 - Justification: Enables management of products/categories without code changes.

Modules [Frontend web pages]

- **CO1 — Front-End Architecture & Framework Design**
 - Module 2: Category Navigator — Declarative UI rendering via `showPage()` + `innerHTML` injection in `app.js`
 - Auth (`auth.js`) — Component-driven login/logout lifecycle, unidirectional data flow via `localStorage`
- **CO2 — JavaScript ES6+ & Modular Architecture**
 - Module 4: Search Bar — `liveSearch()` uses closures, ES6 array methods (`filter`, `map`, `sort`), functional patterns
 - Module 9: Browsing History — `browseHistory` array in `state.js` uses closure as memory/state construct
 - `data.js` — Modular ES6 data layer; `getUserOrders()` uses functional composition + `localStorage` async pattern
- **CO3 — React Component Model & Controlled UI**
 - Module 1: Product Grid — `productCardWithCompare()` is a deterministic UI function: same input = same output
 - Module 7: Product Detail — `renderProductDetail()` with controlled size selector buttons (state via DOM)
 - Tailwind CSS (`index.html`) — Engineering reasoning: speed, consistency, scalability over CSS Modules
- **CO4 — State Architecture & Data Engineering**
 - Module 3: Filter Panel — `state.js` holds global filter state; `toggleFilter()` lifts state across components
 - Module 5: Sort Options — `currentSort` as derived state; co-located with filter rendering pipeline
 - Module 8: Wishlist — `addToWish()` manages global array; data persists via `localStorage` in `getUserOrders()`
- **CO5 — Routing, Forms, Accessibility & Performance**
 - Module 6: Compare Tool — `renderCompareBar()` uses lazy rendering (hidden until items added): performance pattern
 - Checkout (`renderCheckout`) — Controlled form with validation pipeline; `placeOrder()` as error surface handler
 - SPA navigation via `showPage()` in `app.js` — rendering boundaries; protected admin route (`isAdmin()` guard)
- **CO6 — Build Systems, Testing & Deployment**
 - Module 10: Admin Panel — `isAdmin()` protected route; `renderAdminPanel()` with KPI dashboard + product table
 - React wrapper (`App.js`) — CRA build system, npm bundling, script loading order; deployed via React SPA
 - `sdk.js` — Production guard (`if window.elementSdk`); engineering practice for environment-safe deployment

Expected Outcomes of the Proposed Work



Conclusion and Future Scope

- ## Conclusion

- The E-Commerce Product Catalogue successfully integrates frontend, backend, and database technologies to deliver a responsive and functional platform. The project demonstrates practical implementation of modern web development concepts while providing features commonly found in real-world e-commerce applications.

- ## Future Scope

- Future enhancements may include payment gateway integration, AI-based product recommendations, Progressive Web App support, multi-vendor functionality, and dedicated mobile applications for Android and iOS platforms.
- These paragraphs are short enough for presentation slides while still looking professional and not empty.

References

- 1. Mozilla Developer Network (MDN) - HTML5, CSS3, JS Documentation
 - <https://developer.mozilla.org>
- 2. React.js Official Documentation - React Router, Hooks, Context API
 - <https://react.dev>
- 3. Node.js & Express.js Documentation
 - <https://nodejs.org> | <https://expressjs.com>
- 4. MySQL Documentation - Database Design and Querying
 - <https://dev.mysql.com/doc/>
- 5. Axios Library - HTTP Requests in React
 - <https://axios-http.com/docs/intro>
- 6. Bootstrap / Tailwind CSS - Responsive UI Components
 - <https://getbootstrap.com> | <https://tailwindcss.com>
- 7. Postman - API Testing and Development
 - <https://learning.postman.com/docs/>

Thank you